Chapter 7: Stochastic Difference Equations and Moments

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7.1 Basics on random variables

- A probabilistic experiment has a possibility space of chance outcomes.
- A random variable X maps each outcome into a number; its possible values form the range.
- The distribution function maps the range into probabilities.

$$F(x) = \text{Prob}(X \leq x).$$

- $F(x)$ gives the probability that the realization of X is x or smaller.

7.1 Basics on random variables: Discrete vs. Continuous

- If X has a discrete range, $f(x)$ is the probability mass function.

$$f(x) = \text{Prob}(X = x).$$

- If X is continuous, the derivative of F is the density function.

$$f(x) = \frac{dF(x)}{dx}.$$

$$\text{Prob}(a \leq X \leq b) = F(b) - F(a) = \int_a^b f(x)\,dx, \qquad \text{Prob}(X = a) = 0.$$

7.1 Basics on random variables: Illustration

- Tossing a coin twice: possibility space $\{HH, HT, TH, TT\}$.
- Let X be the number of heads, with range $\{0, 1, 2\}$.

$$f(x) = \begin{cases} .25, & x = 0, \\ .5, & x = 1, \\ .25, & x = 2. \end{cases}$$

- For a continuous duration $T \in [0, 4]$, the probability of spending between 1.5 and 2 hours is

$$\text{Prob}(1.5 \leq T \leq 2) = \int_{1.5}^{2} f(t)\, dt, \qquad f(t) = \frac{dF(t)}{dt}.$$

7.2 Examples for random variables: Discrete

- A random variable is characterized by its range and its probability function.
- Discrete uniform distribution:

$$x \in \{a, a+1, \ldots, b-1, b\}, \qquad f(x) = \frac{1}{b-a+1}.$$

- Poisson distribution:

$$x \in \{0, 1, 2, \ldots\}, \qquad f(x) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad \lambda > 0.$$

7.2 Examples for random variables: Continuous

- Normal distribution:

$$x \in (-\infty, +\infty), \qquad f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{ -\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right\}.$$

- Standard normal: $\mu = 0$ and $\sigma = 1$.
- Exponential distribution:

$$x \in [0, \infty), \qquad f(x) = \lambda e^{-\lambda x}, \qquad \lambda > 0.$$

7.2 Examples for random variables: Higher-dimensional

- Consider two random variables X_1 and X_2 that are jointly normally distributed.

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left(\tilde{x}_1^2 - 2\rho\tilde{x}_1\tilde{x}_2 + \tilde{x}_2^2\right)\right\}, \tag{7.2.1}$$

$$\tilde{x}_i = \frac{x_i - \mu_i}{\sigma_i}, \qquad \rho = \frac{\sigma_{12}}{\sigma_1\sigma_2}. \tag{7.2.2}$$

- ρ is the correlation coefficient between X_1 and X_2.

7.2 Examples for random variables: Marginals

- In the two-dimensional normal distribution, each marginal density has the standard form.

$$f(x_i) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left\{-\frac{1}{2}\tilde{x}_i^2\right\}. \tag{7.2.3}$$

- The marginal density is independent of the correlation coefficient ρ.
- Independent normal random variables: impose (7.2.3) for each X_i and set $\rho = 0$.
- Dependent normal random variables: keep (7.2.3) for each X_i and choose $\rho \neq 0$.

7.3 Expected values, variances, covariances and all that

- For continuous random variables, integrals are over the relevant range.
- For discrete random variables, replace integrals by sums.

$$\text{Mean:} \qquad \mathbb{E}X \equiv \int x f(x)\, dx,$$

$$\text{Transformation:} \qquad z = g(x,y) \quad \Rightarrow \quad \mathbb{E}Z = \int\int g(x,y) f(x,y)\, dx\, dy,$$

$$\text{Variance:} \qquad \operatorname{var} X \equiv \int (x - \mathbb{E}X)^2 f(x)\, dx.$$

7.3 Expected values, variances, covariances and all that: Moments

$$k\text{th uncentered moment:} \qquad \mathbb{E}X^k \equiv \int x^k f(x)\,dx,$$

$$k\text{th centered moment:} \qquad \mathbb{E}(X - \mathbb{E}X)^k \equiv \int (x - \mathbb{E}X)^k f(x)\,dx,$$

$$\operatorname{cov}(X, Y) \equiv \int\int (x - \mathbb{E}X)(y - \mathbb{E}Y) f(x, y)\,dx\,dy,$$

$$\rho_{XY} \equiv \frac{\operatorname{cov}(X, Y)}{\sqrt{\operatorname{var} X \operatorname{var} Y}}.$$

- Independence means probability factorization for all relevant events.

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B).$$

7.3 Expected values, variances, covariances and all that: Expectations

- Linearity of expectations is used repeatedly in stochastic difference equations.

$$\begin{aligned}\mathbb{E}[a + bX] &= a + b\mathbb{E}X,\\ \mathbb{E}[bX + cY] &= b\mathbb{E}X + c\mathbb{E}Y,\\ \mathbb{E}(XY) &= \mathbb{E}X\mathbb{E}Y + \mathrm{cov}(X, Y).\end{aligned}$$

- The last identity follows from the covariance definition.

7.3 Expected values, variances, covariances and all that: Variances

$$\begin{aligned}\operatorname{var} X &= \mathbb{E}\left[(X - \mathbb{E}X)^2\right] \\ &= \mathbb{E}\left[X^2 - 2X\mathbb{E}X + (\mathbb{E}X)^2\right] \\ &= \mathbb{E}X^2 - 2(\mathbb{E}X)^2 + (\mathbb{E}X)^2 = \mathbb{E}X^2 - (\mathbb{E}X)^2. \end{aligned} \tag{7.3.1}$$

$$\begin{aligned}\operatorname{var}(a + bX) &= b^2 \operatorname{var} X, \\ \operatorname{var}(X + Y) &= \operatorname{var} X + \operatorname{var} Y + 2\operatorname{cov}(X, Y). \end{aligned} \tag{7.3.2}$$

7.3 Expected values, variances, covariances and all that: Covariances

$$\begin{aligned} \operatorname{cov}(X, X) &= \operatorname{var} X, \\ \operatorname{cov}(X, Y) &= \mathbb{E}(XY) - \mathbb{E}X\mathbb{E}Y, \\ \operatorname{cov}(a + bX, c + dY) &= bd \operatorname{cov}(X, Y). \end{aligned} \tag{7.3.3}$$

- Covariance records whether two variables move together around their means.
- The sign of covariance is central for pro- and countercyclicality.

7.3 Expected values, variances, covariances and all that: Theorem 7.3.1

Theorem 7.3.1

Let X be a random variable and $f(X)$ and $g(X)$ two functions such that

$$f'(X)g'(X) \geq 0 \quad \forall x \in X.$$

Then

$$\text{cov}(f(X), g(X)) \geq 0.$$

- If two variables depend on the same random variable in the same direction, their covariance is positive.
- Example: if both GDP and R&D expenditure rise with TFP, they are procyclical.

7.3 Expected values, variances, covariances and all that: Proof

- Use $Y = f(X)$ and $Z = g(X)$ in the covariance identity.

$$\begin{aligned}\operatorname{cov}(f(X), g(X)) &= \mathbb{E}(f(X)g(X)) - \mathbb{E}f(X)\mathbb{E}g(X) \\ &= \int f(x)g(x)p(x)\,dx - \left[\int f(x)p(x)\,dx\right]\left[\int g(x)p(x)\,dx\right].\end{aligned}$$

- Hence the positive-covariance claim is equivalent to

$$\int f(x)g(x)p(x)\,dx > \left[\int f(x)p(x)\,dx\right]\left[\int g(x)p(x)\,dx\right],$$

- which holds when $f'(X)g'(X) > 0$ by the Chebyshev inequality.

7.3 Expected values, variances, covariances and all that: Linear transformation

- Let $X \sim N(\mu, \sigma^2)$ and define $Y = a + bX$.
- For any random variable,

$$\mathbb{E}(a + bX) = a + b\mathbb{E}X, \qquad \text{var}(a + bX) = b^2 \,\text{var}(X).$$

- A linear transformation of a normally distributed random variable is normally distributed.

$$Y \sim N(a + b\mu, b^2\sigma^2).$$

7.3 Expected values, variances, covariances and all that: Lognormal

- Let $Y = e^X$ where X is normally distributed.
- Then Y is lognormally distributed: $\ln Y \sim N(\mu, \sigma^2)$.

$$\mu_y = e^{\mu + \frac{1}{2}\sigma^2}, \qquad \sigma_y^2 = e^{2\mu+\sigma^2}\left(e^{\sigma^2} - 1\right). \tag{7.3.4}$$

- Since $Y = e^X$, realizations of Y are non-negative.

7.3 Expected values, variances, covariances and all that: Lognormal transformations

- Let Y and Z be lognormally distributed variables.
- Write $Y = e^{X_1}$ and $Z = e^{X_2}$, where the X_i are jointly normally distributed.

$$Y^\alpha = e^{\alpha X_1},$$
$$YZ = e^{X_1} e^{X_2} = e^{X_1 + X_2}.$$

- Since αX_1 and $X_1 + X_2$ are normally distributed, Y^α and YZ are lognormally distributed.
- Total factor productivity is sometimes modelled as lognormal; its logarithm is then normal.

7.3 Expected values, variances, covariances and all that: Theorem 7.3.2

Theorem 7.3.2

Let X be a random variable with density $f(x)$ and range $[a, b]$, which can be $(-\infty, +\infty)$. Let Y be defined by the monotonically increasing function $y = y(x)$. Then the density of Y is

$$g(y) = f(x(y))\frac{dx}{dy},$$

on the range $[y(a), y(b)]$.

- The theorem gives the density of a monotone transformation of a random variable.

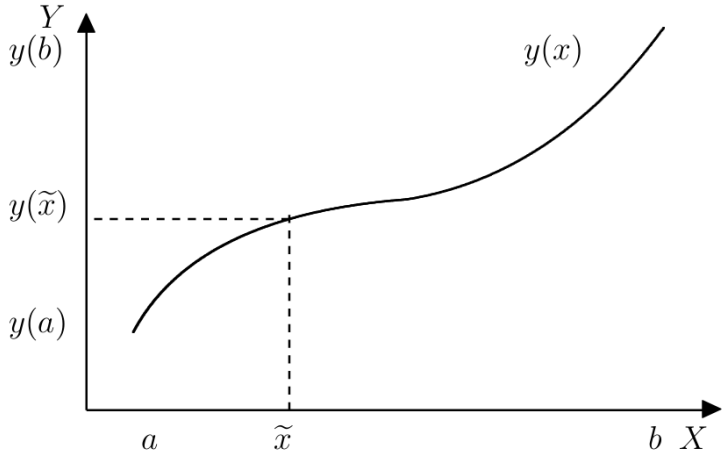


Figure 7.3.1 *Transforming a random variable*

7.3 Expected values, variances, covariances and all that: Transformation proof

- Monotonicity implies that $Y \leq y(\tilde{x})$ if and only if $X \leq \tilde{x}$.
- The range of Y is $[y(a), y(b)]$.

$$
\begin{aligned}
G(y) &= F(x), \qquad y = y(x), \\
G(y) &= F(x(y)), \\
g(y) &\equiv \frac{d}{dy} G(y) = \frac{d}{dy} F(x(y)) = f(x(y)) \frac{dx}{dy}.
\end{aligned}
$$

7.4 Examples of stochastic difference equations

- The first main objective is to describe stochastic processes through stochastic difference equations.
- A simple stochastic difference equation is

$$x_t = ax_{t-1} + \varepsilon_t, \tag{7.4.1}$$

- $a > 0$ is constant, and the stochastic component has mean and variance

$$\varepsilon_t \sim (\mu, \sigma^2).$$

- The shocks are i.i.d.; hence $\text{cov}(\varepsilon_t, \varepsilon_s) = 0$ for all $t \neq s$.

7.4 Examples of stochastic difference equations: Alternative form

- Equation (7.4.1) imposes no specific distribution for ε_t beyond mean and variance.
- The same distributional properties can be represented as

$$x_t = ax_{t-1} + \mu + v_t, \qquad v_t \sim (0, \sigma^2).$$

- Given a boundary condition x_0, repeated substitution solves x_t as a function of time and shock realizations.

7.4 Examples of stochastic difference equations: Substitution

$$
\begin{aligned}
x_1 &= ax_0 + \varepsilon_1, \\
x_2 &= a[ax_0 + \varepsilon_1] + \varepsilon_2 = a^2 x_0 + a\varepsilon_1 + \varepsilon_2, \\
x_3 &= a\left(a^2 x_0 + a\varepsilon_1 + \varepsilon_2\right) + \varepsilon_3 = a^3 x_0 + a^2\varepsilon_1 + a\varepsilon_2 + \varepsilon_3.
\end{aligned}
$$

- Continuing this substitution gives

$$
x_t = a^t x_0 + a^{t-1}\varepsilon_1 + a^{t-2}\varepsilon_2 + \cdots + \varepsilon_t = a^t x_0 + \sum_{s=1}^{t} a^{t-s}\varepsilon_s. \tag{7.4.2}
$$

7.4 Examples of stochastic difference equations: Mean

- The stochastic difference equation predicts the distribution of x_t, not its realized path.
- Define the conditional expectation in period 0 as

$$\tilde{x}_t \equiv E_0 x_t. \tag{7.4.3}$$

- E_0 uses information available at $t = 0$: x_0 is known, but future shocks are unknown.
- Applying E_0 to (7.4.1) yields a deterministic difference equation:

$$\tilde{x}_t = a\tilde{x}_{t-1} + \mu.$$

7.4 Examples of stochastic difference equations: Mean solution

- The expected value evolves according to a deterministic first-order difference equation.

$$\begin{aligned}\tilde{x}_t &= a^t x_0 + \mu \sum_{s=1}^{t} a^{t-s} \\ &= a^t x_0 + \mu \frac{1-a^t}{1-a}, \end{aligned} \tag{7.4.4}$$

- The sum uses

$$\sum_{s=1}^{t} a^{t-s} = a^0 + a^1 + \cdots + a^{t-1} = \sum_{i=0}^{t-1} a^i = \frac{1-a^t}{1-a}.$$

7.4 Examples of stochastic difference equations: Variance

- Start from (7.4.2); $a^t x_0$ is known at $t = 0$ and has zero variance.
- The shocks are i.i.d., so covariances across distinct shocks are zero.

$$\begin{aligned}\operatorname{var}(x_t) &= 0 + \sum_{s=1}^{t} \left(a^{t-s}\right)^2 \operatorname{var}(\varepsilon_s) \\ &= \sigma^2 \sum_{s=1}^{t} \left(a^2\right)^{t-s} = \sigma^2 \sum_{i=0}^{t-1} \left(a^2\right)^i \\ &= \sigma^2 \frac{1 - a^{2t}}{1 - a^2}. \end{aligned} \tag{7.4.5}$$

- Equivalently, this is the conditional variance $\operatorname{var}_0(x_t)$.

7.4 Examples of stochastic difference equations: Long-run behaviour

- In a stochastic environment, the counterpart of a fixed point is a limiting distribution.
- If $\varepsilon_t \sim N(\mu, \sigma^2)$, then every x_t is normally distributed.
- For $0 < a < 1$, the mean and variance converge:

$$\lim_{t\to\infty} \tilde{x}_t \equiv \tilde{x}_\infty = \frac{\mu}{1-a}, \qquad 0 < a < 1.$$

$$\lim_{t\to\infty} \operatorname{var}(x_t) = \frac{\sigma^2}{1-a^2}.$$

- The long-run distribution is normal with these two limiting moments.

7.4 Examples of stochastic difference equations: Distribution of x_t

- Under normal shocks, x_t is normally distributed at every date.
- Its density is pinned down by the time-varying mean and variance.

$$f(x_t) = \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left\{-\frac{1}{2}\left(\frac{x_t - \mu_t}{\sigma_t}\right)^2\right\}.$$

$$\mu_t = \tilde{x}_t \quad \text{from (7.4.4)}, \qquad \sigma_t^2 = \text{var}(x_t) \quad \text{from (7.4.5)}.$$

- With non-normal shocks, (7.4.4) and (7.4.5) remain valid, but the full distribution of x_t is generally not available in closed form.

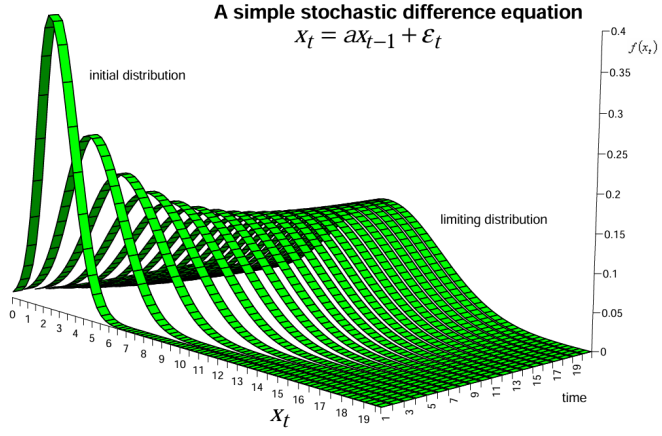


Figure 7.4.1 *Evolution of a distribution over time*

7.4 Examples of stochastic difference equations: Many agents

- Interpret x_{it} as financial wealth of individual i:

$$x_{it} = a x_{i,t-1} + \varepsilon_{it}.$$

- Given $t = 0$ information, the equation predicts the future density $f(x_{it})$ for any individual.
- Shocks are i.i.d. over time and across individuals.
- With a continuum of agents of mass n, idiosyncratic risk averages out.

$$x_t = n \int_{-\infty}^{\infty} x_{it} f(x_{it})\, dx_{it} = n\mu_t, \tag{7.4.6}$$

- Total wealth is deterministic in this pure idiosyncratic-risk setup.

7.4 Examples of stochastic difference equations: Law of large numbers

- Let $P(x_{it} \leq \bar{x}) = \bar{p}$ and define

$$Y_i \equiv I(x_{it} \leq \bar{x}).$$

- Then Y_i is Bernoulli($\bar{p}$) distributed:

$$EY_i = \bar{p}, \qquad \text{var } Y_i = \bar{p}(1 - \bar{p}).$$

- The share of agents below $\bar{x}$ is

$$\bar{q} = \frac{1}{n} \sum_{i=1}^{n} Y_i, \qquad E\bar{q} = \bar{p}, \qquad \text{var } \bar{q} = \frac{\bar{p}(1 - \bar{p})}{n}.$$

- Hence $\lim_{n \to \infty} \text{var}(\bar{q}) = 0$.

7.4 Examples of stochastic difference equations: A more general case

- Now let the coefficient itself be stochastic:

$$x_t = \alpha_t x_{t-1} + \varepsilon_t,$$
$$\alpha_t \sim (a, \sigma_a^2), \qquad \varepsilon_t \sim (\mu, \sigma_\varepsilon^2).$$

- The properties of the process are more complex, but moments can be studied.
- For the limiting distribution, Vervaat's moment expression is

$$Ex^j = \sum_{k=0}^{j} \binom{j}{k} E\left(\alpha^k \varepsilon^{j-k}\right) Ex^k.$$

7.4 Examples of stochastic difference equations: Expected value

- For $j = 1$, the moment expression gives

$$\begin{aligned} Ex^1 &= \sum_{k=0}^{1} \binom{1}{k} E\left(\alpha^k \varepsilon^{1-k}\right) Ex^k \\ &= \binom{1}{0} E(\alpha^0 \varepsilon^1) Ex^0 + \binom{1}{1} E(\alpha^1 \varepsilon^0) Ex^1 \\ &= E\varepsilon + E\alpha \, Ex. \end{aligned}$$

- Solving for the expected value yields

$$Ex = \frac{E\varepsilon}{1 - E\alpha} = \frac{\mu}{1 - a}.$$